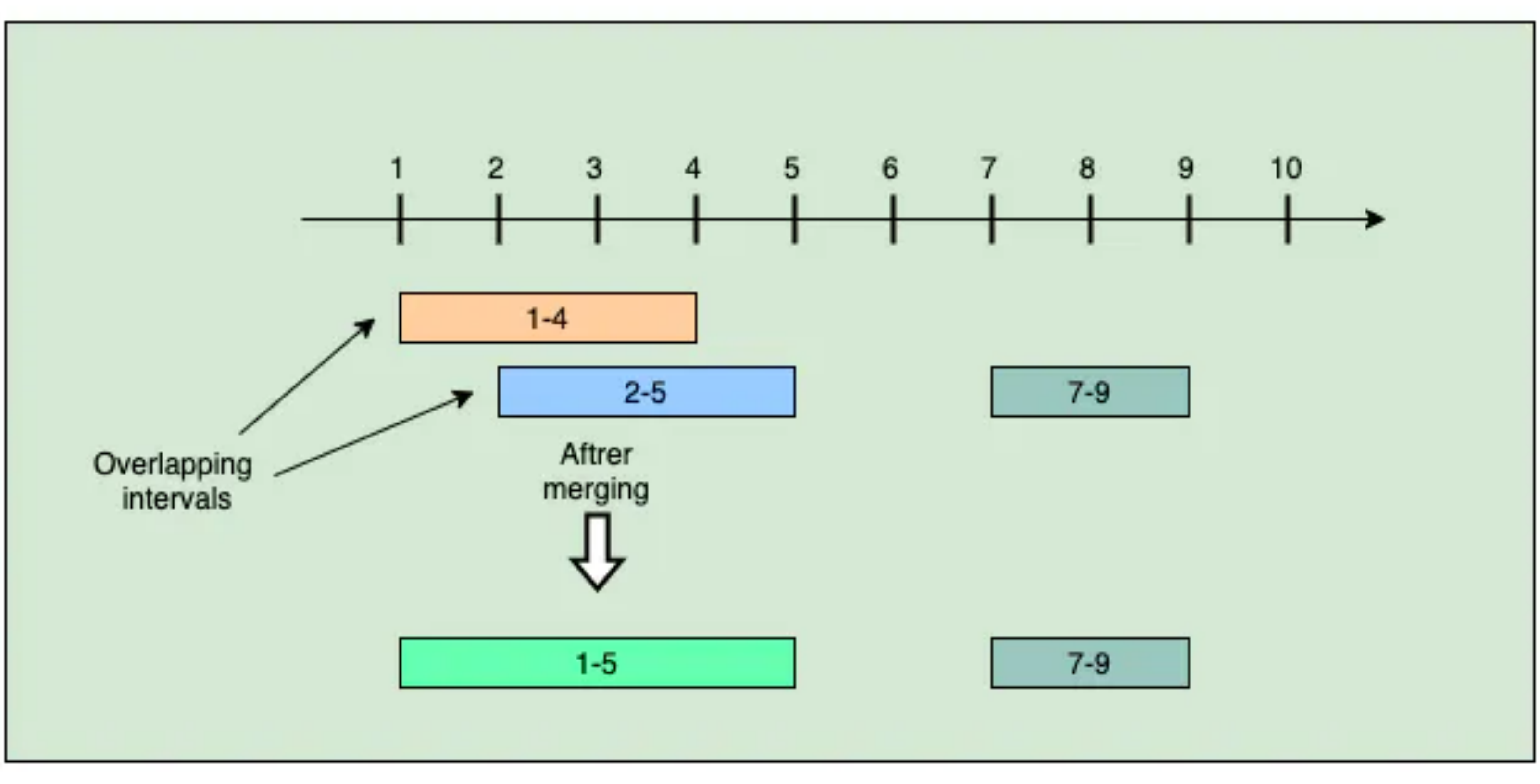
# Solution: Merge Intervals

## Problem Statement

Given a list of intervals, **merge all the overlapping intervals** to produce a list that has only mutually exclusive intervals.

**Example 1:**

```
Intervals: [[1,4], [2,5], [7,9]]
Output: [[1,5], [7,9]]
Explanation: Since the first two intervals [1,4] and [2,5] overlap, we
merged them into one [1,5].
```



**Example 2:**

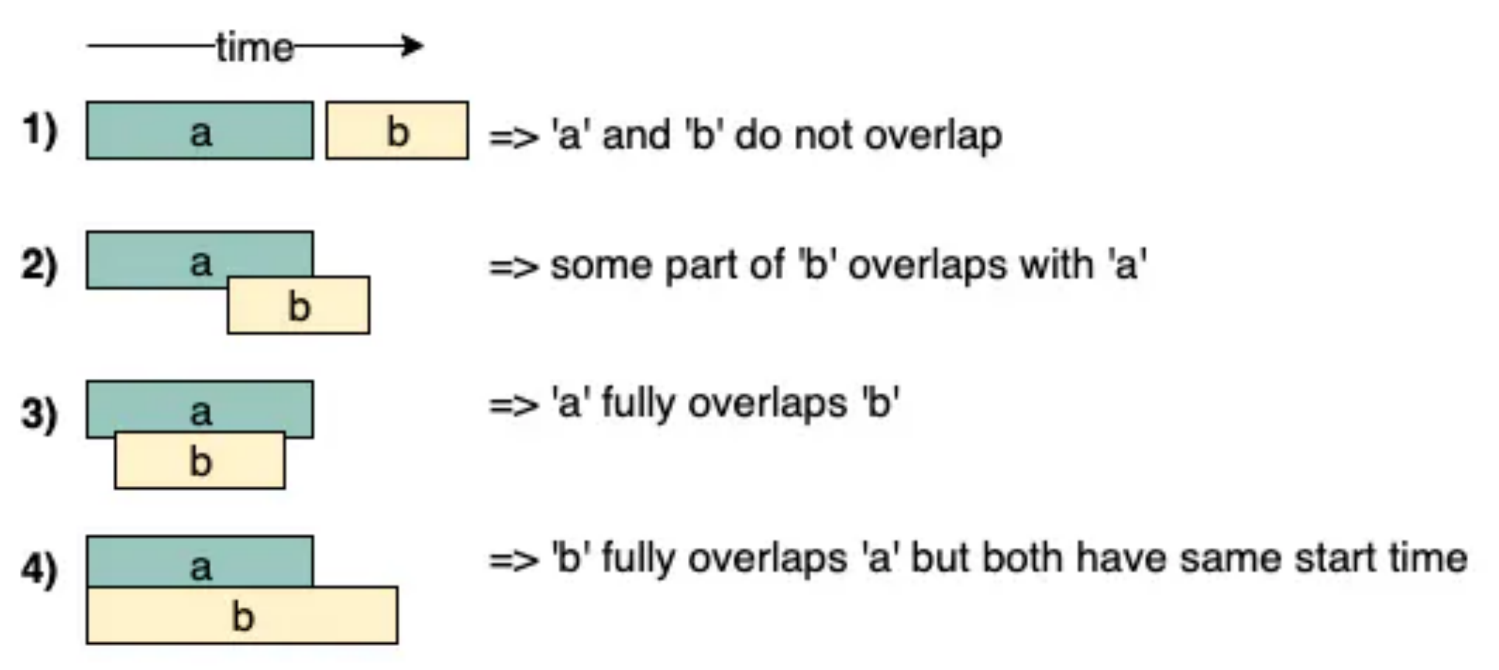
```
Intervals: [[6,7], [2,4], [5,9]]
Output: [[2,4], [5,9]]
Explanation: Since the intervals [6,7] and [5,9] overlap, we merged them
into one [5,9].
```

**Example 3:**

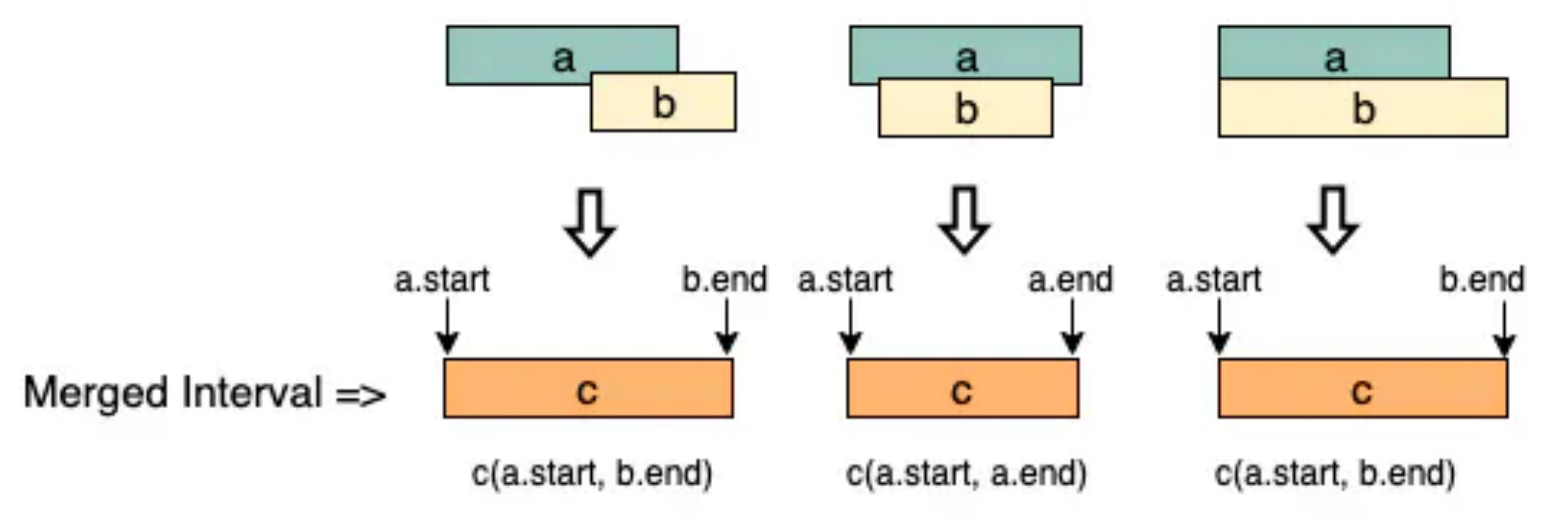
```
Intervals: [[1,4], [2,6], [3,5]]
Output: [[1,6]]
Explanation: Since all the given intervals overlap, we merged them into
one.
```

## Solution

Let's take the example of two intervals ('a' and 'b') such that `a.start <= b.start`. There are four possible scenarios:


time
1)
a
b
=> 'a' and 'b' do not overlap
2)
a
b
=> some part of 'b' overlaps with 'a'
3)
a
b
=> 'a' fully overlaps 'b'
4)
a
b
=> 'b' fully overlaps 'a' but both have same start time

Our goal is to merge the intervals whenever they overlap. For the above-mentioned three overlapping scenarios (2, 3, and 4), this is how we will merge them:



The diagram above clearly shows a merging approach. Our algorithm will look like this:

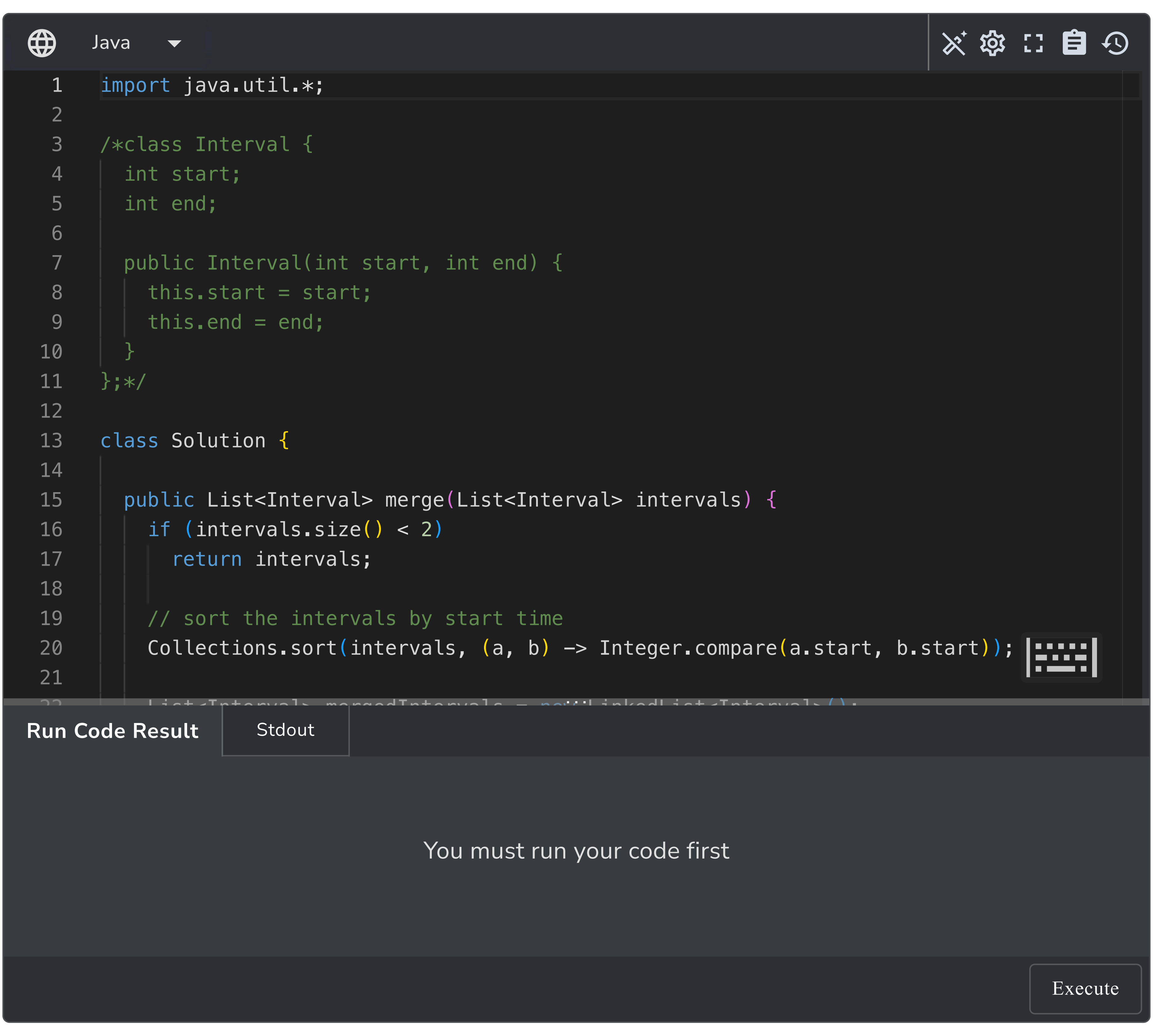
1. Sort the intervals on the start time to ensure `a.start <= b.start`.
2. If 'a' overlaps 'b' (i.e. `b.start <= a.end`), we need to merge them into a new interval 'c' such that:

```
c.start = a.start
c.end = max(a.end, b.end)
```

3. We will keep repeating the above two steps to merge 'c' with the next interval if it overlaps with 'c'.

## Code

Here is what our algorithm will look like:



```java
import java.util.*;

/*class Interval {
  int start;
  int end;

  public Interval(int start, int end) {
    this.start = start;
    this.end = end;
  }
};*/

class Solution {

  public List<Interval> merge(List<Interval> intervals) {
    if (intervals.size() < 2)
      return intervals;

    // sort the intervals by start time
    Collections.sort(intervals, (a, b) -> Integer.compare(a.start, b.start));

```

You must run your code first

### Time Complexity

The time complexity of the above algorithm is $O(N * logN)$, where 'N' is the total number of intervals. We are iterating the intervals only once which will take O(N), in the beginning though, since we need to sort the intervals, our algorithm will take $O(N * logN)$.

### Space Complexity

The space complexity of the above algorithm will be O(N) as we need to return a list containing all the merged intervals. We will also need O(N) space for sorting. For Java, depending on its version, Collections.sort() either uses Merge sort or Timsort, and both these algorithms need O(N) space. Overall, our algorithm has a space complexity of O(N).

## Similar Problems

**Problem 1:** Given a set of intervals, find out if any two intervals overlap.

**Example:**

Intervals: [[1,4], [2,5], [7,9]]
Output: true
Explanation: Intervals [1,4] and [2,5] overlap

**Solution:** We can follow the same approach as discussed above to find if any two intervals overlap.

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